

Umut Onur Yaşar

Applied AI Research Engineer

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Summary

Research engineer working on computer vision and vision–language models, with a background in model efficiency (knowledge distillation, parameter-efficient fine-tuning, reduced-precision inference). I build models and training pipelines from scratch, run controlled multi-seed experiments, and report results with their limitations, including when they contradict the starting hypothesis.

Research

Student Capacity Moderates Knowledge Distillation Effectiveness

Jul 2026

First author, *arXiv preprint (v2; v1 May 2026)*

arxiv.org/abs/2605.31191 · github.com/umutonuryasar/kd-capacity-gap

Logit- versus feature-based KD on CIFAR-10 across four ResNet teacher–student pairs, under a revised protocol with fixed splits, leakage-free model selection and five seeds.

- KD gains were significant for higher-capacity students and indistinguishable from zero for ResNet-18; larger teachers at matched accuracy did not increase gains. Feature-KD matched or exceeded Logit-KD.
- A controlled re-run traced the v1 discrepancy to test-set model selection; the capacity pattern held under the corrected protocol at reduced magnitude.

Selected Projects

MicroCLIP: from-scratch CLIP-style vision–language model

Jul – Sep 2026

Independent project

github.com/umutonuryasar/microclip

Dual encoder built from scratch (ResNet-18 image encoder, 4-layer text transformer, custom 16K BPE tokenizer), trained contrastively on COCO Captions; evaluated on COCO retrieval (R@K) and CIFAR zero-shot.

- Tested whether SigLIP-style sigmoid loss beats softmax InfoNCE at small batch sizes. Softmax matched or outperformed sigmoid at every batch size tested, and the gap widened as batch size shrank. Reported as a small-scale non-replication, with limitations stated.
- Batch 128 and 512 with three seeds each (256 single-seed), plus ablations over optimizer, learning-rate schedule, initialization and a ViT-Tiny encoder; 11 configurations on Colab A100.
- Resumable training for preemptible compute, unit-tested code, in-browser Hugging Face Space demo with int8 ONNX text encoders, all 20 checkpoints published.

RT-DETR knowledge distillation on COCO

May – Aug 2026

Independent project

github.com/umutonuryasar/rt-detr-kd

Distillation of a 15.9M-parameter RT-DETR student from a self-trained 28.9M-parameter ResNet-50 RT-DETR teacher on a 30K-image COCO subset, using a 4 GB laptop GPU and Colab.

- 11 controlled configurations across five KD method families (eight loss variants, including query-level and stage-adaptive distillation), with a held-out selection split, per-loss weight calibration and three-seed evaluation.
- Best configuration raised student AP from 3.9 to 6.8 on val2017 (three seeds) in this 30K-image regime. A weight-swap control showed the gain came from loss weighting rather than the proposed curriculum direction, and simple index-based query alignment outperformed the Hungarian-matching variant.
- Fixed non-deterministic data ordering, a silent resume bug and an fp32-hardcoded positional embedding that broke fp16 inference. fp16 student on an RTX 3050: 6.2 ms latency, 64 MB peak VRAM, no measurable AP loss. Written up in a three-part blog series.

detrflow: RT-DETR inference pipeline

Jun – Jul 2026

Independent project

github.com/umutonuryasar/detrflow

Reproduced the pretrained RT-DETR baseline (53.1 AP on COCO val2017) and wrapped it in a FastAPI inference service with benchmark scripts and a Hugging Face Space deployment.

Open Source

Release cached CUDA memory in k-bit training preparation

Jul 2026

Hugging Face PEFT, merged PR #3293

github.com/huggingface/peft/pull/3293

`prepare_model_for_kbit_training` left about 1 GB of reserved but unused CUDA memory after its fp16-to-fp32 casts on constrained devices; released the cache after the cast loop.

Python 3.14 compatibility fix

Jun 2026

andrewyng/aisuite, merged PR #319

github.com/andrewyng/aisuite/pull/319

Constrained a dependency away from versions that relied on AST APIs removed in Python 3.14.

Experience

Applied AI Research Engineer

Mar 2025 – present

Independent

Ankara, Türkiye

The research, projects and open-source work above, from experiment design to public code, demos and write-ups. Checkpoints and demos: huggingface.co/umutonuryasar.

AI Trainer (part-time)

May 2024 – Feb 2025

Outlier

Remote

Evaluated and ranked LLM outputs for RLHF data pipelines; wrote and reviewed prompts, reference answers and technical evaluations in programming, mathematics and engineering.

Freelance Software Engineer

Sep 2023 – Feb 2025

Self-employed

Ankara, Türkiye

Signal processing and RF analysis tools for paying clients, from algorithm prototypes to desktop applications; own open-source tools include a C++/Qt signal processing toolbox and a Python RF propagation simulator.

Engineering Intern

2015

HAVELSAN

Ankara, Türkiye

Simulation interface development, ROS research, OpenCV-based image processing.

Avionics Intern

2014

TAI – Turkish Aerospace

Ankara, Türkiye

Education

B.Sc. Electrical & Electronics Engineering

2020

Bolu Abant İzzet Baysal University

Bolu, Türkiye

Skills

Methods	Vision–language models, contrastive learning, detection, knowledge distillation
Efficiency	PEFT/LoRA, int8 quantization, mixed precision, GPU memory profiling
Frameworks	PyTorch, Hugging Face Transformers & PEFT, ONNX, OpenCV, FastAPI
Programming	Python, C/C++, Bash, MATLAB
Tooling	Git, Docker, Linux, unit testing
Languages	Turkish (native), English (full professional proficiency)